# ImaginaBook: Guiding Virtual Child Smiles via Behavioral Cues in XR Picture-Book Storytelling

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**Figure 1: Our picture-book storytelling training system: (a) Storytelling to virtual children with a physical book (there is also a VR version using virtual books), (b) Children's facial expressions and exaggerated nonverbal effects providing feedback on the user's performance, (c) Defining system requirements and improving the system through use in actual high school classes.**

## Abstract

Our XR system aids aspiring caregivers in developing vital picture-book storytelling skills for preschoolers. By evaluating four criteria—voice volume, speaking rate, reading posture, gaze direction—the system scores users' performance in real time. Avatars respond dynamically, smiling and clapping in response to high scores, or showing sadness and disengagement when scores are low. This instant feedback loop encourages users to adapt their reading techniques on the spot, making practice sessions both engaging and effective. With both VR and MR modes, the system accommodates virtual and physical books, respectively. Evaluation confirms its potential as a highly valuable training tool.

## Keywords

Virtual Reality, Mixed Reality, Storytelling Training, Interactive Learning

**ACM Reference Format:**
Wataru Aiura, Masanori Yamada, and Shogo Fukushima. 2025. ImaginaBook: Guiding Virtual Child Smiles via Behavioral Cues in XR Picture-Book Storytelling. In *SIGGRAPH Asia 2025 XR (SA XR '25), December 15–18, 2025, Hong Kong, Hong Kong*. ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/3761667.3761951


*SA XR '25, Hong Kong, Hong Kong*

ACM ISBN 979-8-4007-2168-7/25/12
https://doi.org/10.1145/3761667.3761951

## 1 Introduction

Effective picture-book storytelling is a vital component of early childhood education, nurturing creativity, emotional awareness, and language acquisition [Sha et al. 2024]. This practice demands not only narrative comprehension but also technical skills such as appropriate voice modulation, how to hold the book, and directing eye contact [Omura 2024]. However, providing equal practical training opportunities for all aspiring childcare professionals is challenging. Pre-field training using XR technologies presents an attractive solution.

In collaboration with local high schools specializing in childcare, and through multiple interviews with teachers and students, we clarified the requirements for introducing XR systems into childcare classes, established performance evaluation criteria for storytelling, and defined the feedback requirements [Aiura et al. 2024]. In this demonstration, we present a prototype application that provides feedback not only through numerical performance scores but also through children's expressions, actions, and nonverbal effects in the VR space, supporting embodied learning.

The technical contribution of this system is a method that quantifies essential bodily skills in picture-book storytelling, such as voice modulation, book handling, and gaze direction, using only HMD sensor data. These metrics are mapped in real time to the expressions and behaviors of virtual child avatars, providing natural, intuitive feedback without interrupting storytelling. Importantly, the dual requirement of reducing cognitive load while delivering rich information via avatar expressions has not been addressed in existing studies. Insights from this avatar-mediated feedback framework are significant for designing communication support technologies that ensure intuitiveness and low cognitive load.

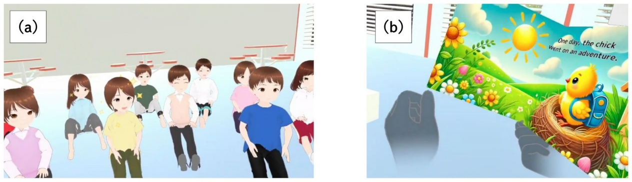


**Figure 2: VR mode: (a)Environment and avatar appearance, (b) manipulating a virtual picture book**

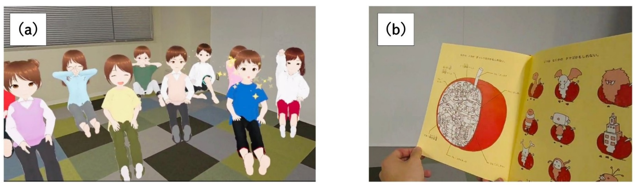


**Figure 3: MR mode: (a)Environment and avatar appearance, (b) manipulating a picture book**

## 2 Design and Implementation

This system is an immersive storytelling support tool featuring two interactive modes: VR Mode and MR Mode. Both modes are designed to enhance users' expressive and communication skills through interaction with virtual child avatars.

In VR Mode, users manipulate a virtual picture book within a fully immersive environment (Figure 2). The system supports both controllers and hand tracking, enabling natural gestures such as page turning. This setup enhances immersion and promotes embodied learning.

In MR Mode, users read a physical picture book aloud while engaging with virtual child avatars (Figure 3). Real-world actions are seamlessly integrated with digital feedback, allowing intuitive interaction across both physical and virtual spaces.

Based on four behavioral criteria identified through interviews with high school childcare specialists, the system evaluates user performance in real time and calculates individual scores for each criterion.

- Voice volume: Measured via the HMD's microphone to assess the appropriateness of vocal projection.
- Speaking rate: Evaluated using speech recognition by analyzing utterance duration and word count.
- Reading posture: Assessed by tracking the position and orientation of the head and hands, particularly whether the book is appropriately oriented toward the avatars.
- Gaze direction: Estimated from the HMD's orientation to determine the quality of eye contact with the avatars.

During the session, a performance score table can be shown or hidden at the user's discretion. The virtual child avatars then react dynamically to the user's aggregated performance score: high scores trigger positive feedback such as smiling, clapping, and sparkling visual effects, whereas low scores lead to negative reactions such as bored facial expressions or diverted gaze. The system currently supports approximately 20 facial expressions and 30 animation patterns.

## 3 Evaluation

A formative evaluation was conducted with 36 high school students in a childcare and welfare program. After using the system, they completed a questionnaire on avatar design, enjoyment, and perceived usefulness. Avatar impressions were rated on a 5-point Likert scale (1 = dislike, 5 = like), and enjoyment and usefulness on a 7-point scale (1 = strongly disagree, 7 = strongly agree).

The mean score for avatar impressions (5-point scale) was 4.66 (SD = 0.59), indicating generally favorable responses. Enjoyment received a high mean of 6.83 (SD = 0.45), and perceived usefulness was rated at 6.40 (SD = 0.95).

These findings suggest the system was well-received. Positive responses to the avatar design support its effectiveness as a user interface. High enjoyment and usefulness scores indicate strong potential for future use and improvement.

## 4 Discussion and Future Work

The system enables users to elicit smiles from virtual child avatars through picture-book storytelling, demonstrating the potential of XR technologies to foster expressive and communication skills in childcare training. Evaluation results indicated that participants perceived the system as both enjoyable and practically applicable.

Nevertheless, several limitations were identified. First, avatar reactions are currently based on aggregated performance scores, which do not provide criterion-specific feedback. Second, reliance on HMDs and controller-based sensors limits accurate estimation of full-body posture and precise gaze tracking.

Future work should focus on refining the scoring logic and incorporating eye-tracking–enabled HMDs alongside advanced sensing technologies to improve the reliability of assessment. Avatar diversity and the range of emotional expressions should also be enhanced to increase the system's adaptability across various training scenarios. Moreover, the integration of large language models (LLMs) is expected to enable natural conversational interactions with avatars regarding picture-book content, as well as to provide detailed feedback following storytelling sessions. Finally, longitudinal studies will be necessary to evaluate the extent to which training effects observed in the virtual environment translate to real-world childcare practices.

## Acknowledgments

We express special thanks to the Fukuoka City Educational Council and Fukuoka Municipal Fukuoka Girls' High School. This research was supported by JSPS Kaken-hi (JP24834281, JP25H00552) and the JST Program for Co-Creating Startup Ecosystem (JPMJSF2317).

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